# Excel Report Generator

Internship Project Report | Python & Data Analysis

## 1. Introduction

The Excel Report Generator is a Python-based data analysis and reporting application developed as an internship project. It provides a simple Streamlit interface through which users can upload CSV datasets, process the data, view analytical summaries and visualisations, and generate a structured Excel report. The project focuses on reducing repetitive manual work involved in analysing tabular data and preparing reports.


## 2. Abstract

The application automates a basic data-to-report workflow. After a CSV file is uploaded, the system removes completely empty and duplicate rows and, when Quantity and Unit_Price fields are available, calculates Total Sales. It presents record counts, sales metrics, summary statistics, category and regional analysis, and top products through a Streamlit dashboard. The application also creates a downloadable Excel workbook containing a summary sheet, processed data, category analysis, and an Excel chart.


## 3. Tools Used

| Tool / Library | Purpose |
|---|---|
| Python | Core programming language |
| Streamlit | Interactive web-based user interface |
| Pandas | CSV loading, cleaning, transformation and analysis |
| Matplotlib | Data visualisation |
| OpenPyXL | Excel workbook creation, formatting and chart generation |

## 4. Steps Involved in Building the Project

- **Dataset Upload:** A CSV file is uploaded through the Streamlit file uploader.
- **Data Processing:** The dataset is loaded with Pandas, completely empty rows are removed, and duplicate records are removed.
- **Derived Metric:** When Quantity and Unit_Price columns are present, Total Sales is calculated as Quantity × Unit Price.
- **Data Analysis:** The application displays record counts, total sales, average sales, numerical summary statistics and a data preview.
- **Visualisation:** Sales are analysed by category and region using Matplotlib, while product-level sales are used to identify top products.
- **Excel Report:** OpenPyXL creates a workbook containing Summary, Data and Category Analysis sheets, including an Excel bar chart.
- **Download:** The generated workbook is provided through a download button for further use.

## 5. Conclusion

The Excel Report Generator successfully demonstrates how Python can be used to automate data processing, analysis, visualisation and Excel report generation in a single application. The project combines Pandas, Matplotlib, OpenPyXL and Streamlit to provide a practical workflow for turning CSV data into an analytical report. It also provides a clear foundation for future enhancements such as additional datasets, more visualisations and advanced reporting features.